

Roni Singh

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PROFILE

M.E. CSE (AI & ML) student at Jadavpur University | B.Tech (IT) graduate, MAKAUT (CGPA 8.32/10) | GATE 2026 (CS) qualified | Experienced in ML, XAI, and software development. Seeking Software Engineer or ML Engineer opportunities.

CERTIFICATIONS & COMMUNITY

- Qualified GATE 2026 (CS/IT) – one of India's most competitive CS exams, demonstrating rigorous knowledge of algorithms, data structures, OS, DBMS, and computer networks.
- Software Development Intern at CodTech Technology – gained hands-on experience in full-stack web development, UI design, and project collaboration using React, Next.js, and TypeScript.
- Organized and co-led technical events for 100+ participants as Core Member of HACKAUT Tech Club.

TECHNICAL SKILLS

Languages Python, Java, C
ML / Data TensorFlow/Keras, Scikit-learn, NumPy, Pandas, Matplotlib, SHAP, LIME
Backend MySQL, Flask (REST APIs)
Tools Git, GitHub, Jupyter Notebook
Concepts Machine Learning, Deep Learning, XAI, LSTM, CNN, Feature Engineering
Soft Skills Teamwork, Problem Solving, Communication, Leadership, Time Management, Adaptability

PROJECTS

Malware Analysis & Classification – Python · TensorFlow · Scikit-learn · SHAP · LIME · NetworkX · Angr

- End-to-end detection pipeline on Windows PE files – extracting static (entropy, headers), dynamic (API traces), and graph-based (CFG) features for multi-modal classification.
- Trained CNN, Random Forest, and SVM models; achieved **F1 > 0.95** across 8 malware families on held-out test set.
- Integrated SHAP & LIME explainability, making model decisions auditable for security analysts.

Stock Price Prediction with LSTM – Python · TensorFlow · Pandas · Matplotlib

- Stacked LSTM on 5 yr OHLCV data; achieved **RMSE ~2.3%** on NIFTY50 stocks, outperforming ARIMA and linear regression baselines.
- Walk-forward validation eliminates look-ahead bias and ensures realistic out-of-sample metrics.

Digital Product Marketplace – Next.js · React · TypeScript · Tailwind CSS · MongoDB

- Built a full-stack e-commerce platform for selling digital products such as templates, greeting cards, UI kits, and design assets.
- Developed secure authentication, product listings, shopping cart, checkout, and instant download functionality after purchase.
- Implemented responsive UI with dark/light mode, category-based browsing, search, and product detail pages.
- Integrated payment workflow with Card and PayPal options and designed a modern user experience similar to leading digital marketplaces.

Payroll Management System – Python · MySQL · Flask

- Web-based payroll tool automating TDS computation, payslip generation, and batch processing for 50+ employees via a Flask front-end.
- Normalised relational schema (employees, departments, payroll_runs) with stored procedures for monthly runs.

EDUCATION

M.E. – Computer Science & Engineering (AI & ML) | Jadavpur University *July 2026 – Present*

- Currently pursuing M.E. with specialization in Artificial Intelligence and Machine Learning.

B.Tech – Information Technology | Maulana Abul Kalam Azad University of Technology, WB *July 2022 – June 2026*

- SGPA: **8.34 / 10.0** Coursework: DSA, DBMS, Machine Learning, Computer Networks

WBCHSE – Higher Secondary | Burdwan CMS High School *June 2022*

Score: 72%

WBBSE – Secondary | Ramharipur Ramakrishna Mission High School *June 2020*

Score: 86%